TransitShield — IBM Z Datathon 2026 Project Proposal

Theme: Wildcard — AI for Community Impact and Sustainability

1. Executive summary

TransitShield is an AI-powered public transport prediction system designed to help identify bus delays, cancellations and potential service disruptions before they significantly affect passengers. It uses historical transport data and machine learning to predict which routes are at risk and recommend where transport operators should focus their attention.

Public transport disruptions can cause passengers to arrive late to work, miss appointments and experience difficulties connecting to other services. They can also make it harder for people who depend on public transport to plan their day.

TransitShield aims to help transport operators make better-informed decisions by turning transport data into useful predictions and recommendations. The project will use the IBM LinuxONE Community Cloud platform to run its AI and machine learning workloads, demonstrating how secure, scalable enterprise computing can support a practical community problem.

The proposed prototype will include a prediction model, a dashboard displaying route risks and a recommendation system that explains which routes may need attention. The team will evaluate the model using historical data and demonstrate its behaviour using test scenarios.

2. Problem statement

Public transport networks generate data about scheduled services, actual departure times, delays, cancellations and passenger demand. However, identifying patterns that could indicate future disruptions can be difficult when this information is spread across large datasets.

When transport operators cannot identify potential problems early, they may have less time to adjust schedules, communicate with passengers or allocate available resources.

TransitShield will investigate how machine learning can use historical transport data to predict potential bus disruptions and help operators decide where intervention may be most useful.

The project will focus on three questions:

- Can machine learning identify routes and conditions associated with a higher risk of delays or cancellations?

- Can these predictions be presented in a way that helps operators prioritise their attention?
- Can the solution run on IBM LinuxONE Community Cloud while demonstrating the platform's AI, computing and security capabilities?

These questions will guide the prototype and its evaluation. The project will not assume that predictions automatically prevent disruptions; instead, it will test whether the system can provide useful advance warnings.

3. Proposed solution

TransitShield will combine predictive machine learning with a simple dashboard that helps transport operators understand and respond to potential service disruptions.

A. Transport data analysis

The system will process historical service records, including scheduled and actual departure times, cancellation status, route identifiers and dates. Where available, passenger demand and weather information may also be included.

B. Machine learning predictions

A trained model will estimate the likelihood of a delay or cancellation based on patterns in the available data. The team will compare its predictions against actual outcomes to measure performance.

C. Route risk dashboard

A dashboard will display routes with elevated predicted risk, explain the main contributing factors where possible, and help operators identify which services warrant further investigation.

D. Action recommendations

Based on predictions and available operational information, the system will suggest actions such as reviewing a timetable, checking a potentially disrupted service or considering whether spare buses could be allocated. Final decisions remain with transport operators.

4. Target users and community benefits

TransitShield is designed primarily for public transport operators and service planning teams. Passengers are the wider community that could benefit from improved service planning.

Stakeholder	**Potential benefit**

Transport operators	Earlier visibility of potentially disrupted routes
Passengers	Potentially more reliable services and better-informed travel planning
People who rely on public transport	Better identification of services where disruptions could have a significant impact
Local communities	More informed decisions about transport resources and service improvements

The project supports the wildcard theme because it applies AI to a practical community issue rather than using AI for automation alone. It also has potential sustainability benefits: better-informed resource allocation could help reduce unnecessary vehicle deployment, although this benefit would need to be measured rather than assumed.

5. AI and machine learning approach

The team will start with a manageable prediction task and expand it if time permits.

Proposed model workflow

Historical transport records

Scheduled times, actual times, route and service status

Data cleaning and feature engineering

Calculate delay minutes, encode route and time features, handle missing values

Predictive machine learning model

Estimate delay or cancellation risk from information available before the service

Risk scores and recommendations

Rank routes for review and display results in the dashboard

Proposed technical methods

- Data processing: Python and pandas for cleaning and transforming transport records.
- Baseline model: Logistic regression or a decision tree to estimate disruption risk.
- Optional advanced model: Random forest or gradient boosting if the available data supports a meaningful improvement.
- Evaluation: Precision, recall, F1-score and a confusion matrix. Recall is particularly important because missing a real disruption may be costly.

- Visualisation: A dashboard built with a suitable Python web framework and charts showing risk levels and historical trends.

The target outcome could initially be whether a bus service will be significantly delayed or cancelled. The team must define the delay threshold before training and evaluation, based on the dataset and the intended use.

The model must only use information that would be available at prediction time. For example, actual arrival time cannot be used to predict whether that same service will be delayed before it departs. This helps prevent data leakage and makes the evaluation more realistic.

The system will not claim to predict disruptions perfectly. It will be tested against a held-out portion of the dataset, and its limitations will be reported honestly.

6. What makes TransitShield different?

Basic transport dashboards report what has already happened. TransitShield aims to help operators identify what may happen next and understand the possible consequences.

To make the idea more distinctive, the team could develop a feature called TransitShield Ripple.

TransitShield Ripple: Predicting disruption impacts

Instead of assessing each route separately, the system would investigate how a potential disruption could affect connected services and passengers transferring between them.

For example, if a delayed bus connects with another service, TransitShield Ripple could flag the connection as potentially at risk and recommend that an operator review the situation.

This feature depends on having suitable route and timetable data. If those records are unavailable, the team can demonstrate the concept using a small, clearly labelled simulated network.

This feature could give the project a stronger originality angle because it focuses on the wider consequences of disruptions, rather than simply predicting whether an individual bus will be late.

The team should treat the ripple feature as an extension after the core predictive model works. That keeps the project achievable while still leaving room for innovation.

7. IBM Z (LinuxONE) integration and technical architecture

Using IBM LinuxONE Community Cloud is a core requirement of the Datathon, not an optional hosting choice. The judging criteria specifically assess the quality of the architecture, use of AI/ML packages on the platform, use of platform capabilities, and supporting screenshots. Both supplied AI/ML labs are also mandatory prerequisites before building the team's solution.

Pasted markdown.md

The proposed architecture is:

Transport data source

Historical records and permitted simulated data

IBM LinuxONE Community Cloud

Data processing and feature preparation

Machine learning inference and risk scoring

Recommendation service and application API

Containerisation and access controls will be used where supported by the provided environment.

TransitShield dashboard

Route risk rankings, predicted disruption risks, explanations and recommended reviews

How the platform contributes

- AI/ML execution: Train or run the predictive model on LinuxONE, according to the available environment and Datathon requirements.
- Containerisation: Package the application and its dependencies into containers if supported.
- Security: Restrict access to the application and data, and avoid exposing sensitive information unnecessarily.
- Scalability: Design the data-processing and prediction components so they can handle additional routes and records.
- Evidence: Capture genuine screenshots of the running platform, model execution, application components and dashboard.

The proposal does not assume that the team will have access to every LinuxONE capability. The actual implementation must be based on the Community Cloud resources, permissions and tools provided during the event.

Mandatory preparation

Before building TransitShield, the team must complete both supplied labs:

1. Predictive AI Hybrid AI scenarios (multi-model).
2. Predictive AI + Generative AI.

Pasted markdown.md

The team should review the patterns demonstrated in these labs and determine how to apply them to the transport problem. If the environment supports a suitable generative AI model, it could help translate prediction results into short explanations for operators. The predictive model should remain responsible for calculating disruption risk; generated explanations should not invent evidence or replace the underlying predictions.

8. Project objectives and success measures

The following are proposed goals for the prototype, not achieved results.

Objective	**How success will be measured**
Predict disruptions	Evaluate precision, recall and F1-score on held-out data
Provide useful warnings	Measure how many actual disruptions the model identifies and how often its alerts are incorrect
Help operators prioritise	Verify that the dashboard ranks higher-risk services clearly and consistently
Explore ripple effects	Test whether the extension identifies potentially affected connections in a sample network
Use IBM LinuxONE	Demonstrate the application and AI/ML workload running on the required platform
Protect data	Verify access controls and ensure the demo does not expose unnecessary personal information

The team should establish a baseline model first. If the more complex model does not improve meaningful performance measures, the simpler model may be the better choice.

The team should also measure the time between receiving a record and producing a prediction if the environment permits. This will help demonstrate the system's responsiveness without claiming it is a real-time production service before that capability has been tested.

9. Project scope and deliverables

To keep the project achievable during the Datathon, TransitShield will be developed in stages.

1. Working data pipeline: Load, clean and prepare a transport dataset.
2. Predictive model: Train and evaluate a model that identifies potential delays or cancellations.
3. Risk dashboard: Display route risk scores, relevant historical information and suggested actions.
4. LinuxONE deployment: Run the relevant AI/ML workload on IBM LinuxONE Community Cloud and document the implementation.
5. Demonstration: Show a test scenario, explain the prediction, and report the model's actual evaluation results.

The Ripple feature will be included if the core system is completed early enough.

10. Proposed development schedule

The exact schedule should be adjusted to match the Datathon's published timeframe.

Stage	Activities	Expected output
Preparation	Complete both mandatory labs and inspect available data	Platform familiarity and dataset selection
Initial development	Clean data and establish a baseline model	Initial evaluation results
AI development	Improve the model and create risk scores	Tested prediction pipeline
Application development	Build dashboard and recommendations	Working prototype
Deployment	Integrate and test on LinuxONE	Running application and evidence
Final delivery	Test the demo, prepare the pitch and record the video	Competition submission

The team should build a working minimum viable product early, rather than spending too long on model complexity or dashboard appearance before the main prediction works.

11. Risks and limitations

Risk	Proposed response
Incomplete or unsuitable data	Start with a smaller dataset and document missing information
Inaccurate predictions	Compare against a baseline, report false positives and false negatives, and avoid unsupported claims
Data leakage	Ensure every model input would be available at prediction time
Limited development time	Prioritise the core model, dashboard and LinuxONE deployment before Ripple
Platform restrictions	Confirm available packages, deployment methods and resources during preparation
Unnecessary resource recommendations	Present suggestions for human review rather than automatically changing transport services

TransitShield will be a decision-support prototype. It will not control buses, change live timetables or guarantee that passengers avoid delays. Those capabilities would require additional operational data, validation and integration with real transport systems.

12. Team responsibilities

The roles below can be assigned based on the size and skills of the team.

Data and machine learning

Prepare the dataset, build the prediction model, evaluate results and document limitations.

IBM LinuxONE and deployment

Configure the environment, run the AI/ML workload, manage dependencies and collect platform evidence.

Dashboard and application

Build the interface, display predictions, and connect the dashboard to the prediction service.

Testing and presentation

Coordinate test scenarios, verify results, prepare the demonstration and communicate the community benefits.

Team members can share responsibilities, particularly if the team is small. All project code must be created during the Datathon timeframe, in line with the originality requirements in the judging criteria.

Pasted markdown.md

13. How TransitShield addresses the judging criteria

The supplied rubric evaluates community value and theme alignment, originality, technical design and LinuxONE/AI integration, and the quality of the final delivery.

Pasted markdown.md

Judging category	TransitShield strategy
Community value and theme alignment	Address transport reliability and show how better predictions could support passengers and operators
Novelty and originality	Develop the Ripple extension to explore potential effects on connected services
Technical design	Run the predictive AI/ML workload on LinuxONE and demonstrate platform capabilities with genuine evidence
Execution and pitch	Deliver a working end-to-end demonstration supported by measured results and a clear explanation

For a stronger entry, the team should focus on demonstrating what makes TransitShield useful and technically credible, rather than simply listing features. The additional AI Innovation Excellence rubric places particular emphasis on originality and technical sophistication, so the Ripple concept could be valuable if the core solution is already working.

Pasted markdown.md

14. Proposed final demonstration

The presentation video could follow this sequence:

1. Introduce the problem. Explain how bus disruptions affect passengers and transport operators.
2. Show the data. Briefly demonstrate the historical records used by the model.
3. Run a prediction. Enter a test scenario and show the predicted disruption risk.
4. Explain the result. Show which available features contributed to the prediction and why the route may need attention.

5. Demonstrate the dashboard. Display the risk ranking and proposed operator actions.
6. Show LinuxONE integration. Provide genuine evidence of the application and AI/ML workload running on the required platform.
7. Present the evaluation. Report the actual model metrics, limitations and any observed benefits from testing.

The final video should follow the Datathon's published time limit. No performance figures or deployment outcomes should be included until the team has measured them.

15. Conclusion

TransitShield proposes a practical use of AI to help public transport operators anticipate potential bus delays and cancellations. By combining historical transport data, machine learning, risk rankings and actionable recommendations, the system aims to help operators identify problems earlier and make better-informed decisions.

The project will use IBM LinuxONE Community Cloud to demonstrate secure and scalable execution of the AI/ML workload. Its optional Ripple extension will investigate how disruptions on one service may affect connected routes, providing an opportunity to explore a more distinctive approach to transport planning.

The main goal is to build a working, honestly evaluated prototype that demonstrates how AI can turn transport data into useful decisions for the community.